

Logan Kojiro

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SUMMARY

Experienced software engineer with a background building verification and testing platforms for complex hardware systems — officially titled “Verification Engineer,” though the role centered on software tooling rather than SystemVerilog test-bench work. Now focused on real-time computer graphics and GPU-accelerated rendering, with a foundation in C++, CUDA, parallel computing, and linear algebra applied hands-on in a from-scratch 3D rasterizer and a GPU-parallelized fluid simulator. Looking to bring a rigorous, detail-oriented engineering approach to a team building real-time rendering or graphics technology.

WORK EXPERIENCE

Ampere Computing — Verification Engineer

Santa Clara, CA | Aug 2021 – Mar 2026

- Co-built and served as secondary owner of an internal Python coherency-checking tool modeling Ampere’s mesh network against the CHI and AXI protocols — adopted by 90%+ of the mesh DV team and run alongside thousands of weekly regression tests, catching hundreds of bugs in the team’s proprietary hierarchical coherency model.
- Triaged test failures against design specs to separate real RTL design bugs (routed to component owners) from tool bugs (fixed directly); helped scale the tool’s core team from 2 to 4 engineers.
- Architected a C++ DPI layer enabling mid-run bug detection in simulation and emulation, letting engineers kill failing runs early and cut wasted compute on long overnight regressions.
- Contributed to a company-wide quarterly “Kilowatt” award (team of 3) recognizing the tool’s impact on debugging turnaround time.

ManTech — CNO Developer Intern

Herndon, VA | Summer 2019, Summer 2020

- Built a JavaScript browser-fingerprinting tool that identified and enumerated features of unknown browsers with over 90% accuracy, plus a companion suite of Python tools to verify and extend its functionality.
- Wrote position-independent code in Assembly and C to extract target operating system information from within a browser sandbox.

ARIN Technologies — Software Engineer Intern

Pittsburgh, PA | Summer 2018, Spring 2019

- Designed and implemented a scalable, user-friendly GUI that cut calibration time for ARIN’s indoor positioning system by 95% compared to manual calibration.

CMU BioRobotics Lab — Research Assistant

Pittsburgh, PA | Feb 2017 – Aug 2017

- Collaborated with an interdisciplinary team of engineers and biologists to emulate snake locomotion and behavior in a robotic platform.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA | May 2021

B.S., Electrical & Computer Engineering

PROJECTS

Terminal Tracer — C++20 Real-Time 3D Rasterizer

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- Built a from-scratch software rasterizer rendering real-time ASCII terminal graphics from .obj mesh files.
- Implemented Z-buffering, MSAA anti-aliasing, and Lambertian shading for accurate depth handling and surface lighting.

GPU Fluid Simulation — SPH Solver in CUDA, OpenMP & OpenGL

CMU Parallel Computing Course

- Implemented a Smoothed Particle Hydrodynamics fluid solver in three variants — sequential, OpenMP, and CUDA — with real-time rendering through an OpenGL Vertex Buffer Object pipeline.
- Designed a Z-order (Morton code) spatial grid with a GPU-optimized compact variant, bounding each particle’s neighbor search to 27 local blocks instead of a naive $O(N^2)$ scan.
- Diagnosed and resolved a CUDA kernel serialization bug caused by atomic-operation contention during grid construction, achieving up to a 279,654x CUDA speedup and real-time simulation of 100,000+ particles.

TECHNICAL SKILLS

Languages: C, C++, Python, CUDA C, Assembly

Graphics & Rendering: Real-time rasterization, Z-buffering, MSAA anti-aliasing, Lambertian shading, OpenGL, GPU-accelerated rendering, 3D geometry & mesh processing

Systems & Tools: Unix/Linux, Git, OpenMP, Parallel Computing, Design Verification (DV)

LEADERSHIP & ACTIVITIES

Carnegie Mellon ECE Department — Teaching Assistant | Sigma Phi Epsilon Fraternity — Greek Sing Chair, House Manager | Boy Scouts of America — Eagle Scout, Patrol Leader, Instructor